Supplementary Material for A Physiology-Informed Deep Learning Approach for Sleep-EEG Brain Age Estimation

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1 Supplementary Methods

1.1 Probing for sleep-stage information

As part of our sleep-stage-level analysis, we performed a probing experiment to determine whether the embeddings from our phase-1 model contained sleep-stage information. Since each embedding corresponds to a 5-minute window (10 epochs), we first assigned a "dominant stage" label to each one. An embedding was labeled with a specific sleep stage if that stage constituted 60% or more of the corresponding window. If no single stage met this criterion, the embedding was excluded from this analysis.

With these labeled embeddings, we then investigated whether the model had learned stage-related features despite only being trained on age. To test this, we trained a logistic regression classifier to predict the dominant stage labels using the embeddings as input features. The classifier was evaluated using five-fold cross-validation across subjects on the test set. To quantify the stage-specific information captured, we report the sleep-staging accuracy and macro-F1 score averaged across folds.

1.2 Clinical and Demographic Associations With BAG

As an exploratory secondary analysis, we examined associations between BAG and available demographic, sleep, and medical-history variables using multivariable linear regression. BAG was modeled as a continuous outcome in years. Covariates included sex, BMI, AHI, arousal index, and histories of stroke, depression, coronary heart disease (CHD), diabetes, hypertension, and smoking. This analysis was designed to summarize available associations with BAG and was not used for model training or model selection.

1.3 Exploratory per-filter occlusion test as a sanity check

We also performed a sanity check using a straightforward single-filter occlusion test. After the model was trained, we systematically evaluated the importance of each filter in the learned Gabor filter bank. For each filter, we created an "occluded" version of the model by setting that filter's activation output to zero during inference.

While this method simulates the effect of missing information from a single filter, it does not account for potential nonlinear relationships or interactions between filters. Therefore, this analysis served only as a basic sanity check. To quantify the impact of occluding each filter, we computed two metrics: the change in mean absolute error ($\triangle$MAE) between the occluded and full models and the Spearman rank correlation between their predictions to assess rank-order stability. The results are shown in Supplementary Fig. S7.

2 Supplementary Results

2.1 Assessing sleep-stage information in learned embeddings

Given that N2 sleep provided consistently high SHAP values, we tested whether the learned embeddings retained sleep-stage information despite the absence of sleep-stage labels during age-model training. The stage-specific SHAP contributions correlated with the ground-truth percentage of time each individual spent in that stage (Wake: $r =$

0.74, N1: $r = 0.69$, N2: $r = 0.77$, N3: $r = 0.91$, and REM: $r = 0.80$, all $p < .001$) (Supplementary Fig. S6A), indicating that stage-level attributions are related to full-night stage prevalence. The correlations among SHAP values, internal attention scores, and stage percentage indicate concordance among these measures.

A linear probe trained on the model-learned embeddings further supported that the network captured sleep-stage information (Supplementary Fig. S6B). Using five-fold cross-subject validation, a classifier achieved an accuracy of $79.4 \pm 0.18\%$ and a macro-F1 score of 0.59 ± 0.00 when predicting the dominant stage of 5-minute windows. This performance exceeded the 20% chance level for a five-class problem, indicating that the embeddings retain information about sleep stage.

2.2 Clinical and demographic associations with BAG

In exploratory multivariable linear regression of BAG (Supplementary Table S6), chronological age was associated with lower BAG ($\beta = -0.23$ years per year; 95% CI, -0.25 to -0.22). Male sex was associated with higher BAG relative to female sex ($\beta = 2.18$ years; 95% CI, 1.78 to 2.58). Arousal index was also positively associated with BAG ($\beta = 0.69$ years per 10 events/h; 95% CI, 0.49 to 0.89), as were depression history ($\beta = 0.74$ years; 95% CI, 0.01 to 1.46), diabetes history ($\beta = 0.75$ years; 95% CI, 0.15 to 1.36), and hypertension history ($\beta = 0.60$ years; 95% CI, 0.24 to 0.96). Smoking history was associated with lower BAG ($\beta = -1.06$ years; 95% CI, -1.84 to -0.28). BMI, AHI, stroke history, and CHD history were not nominally associated with BAG. These associations are exploratory and should not be interpreted as causal effects or as evidence that the model detects specific diseases.

3 Supplementary Figures

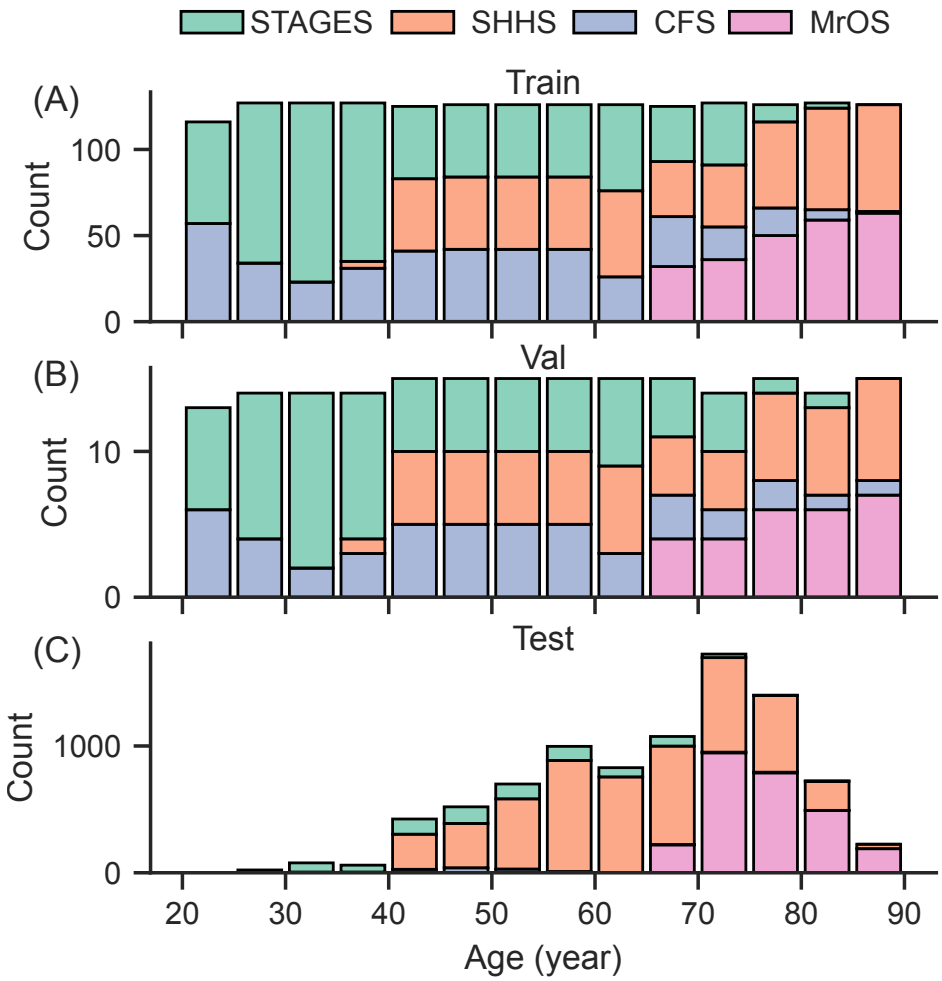


Figure S1: Distribution of age and source cohort across the training, validation, and testing sets.

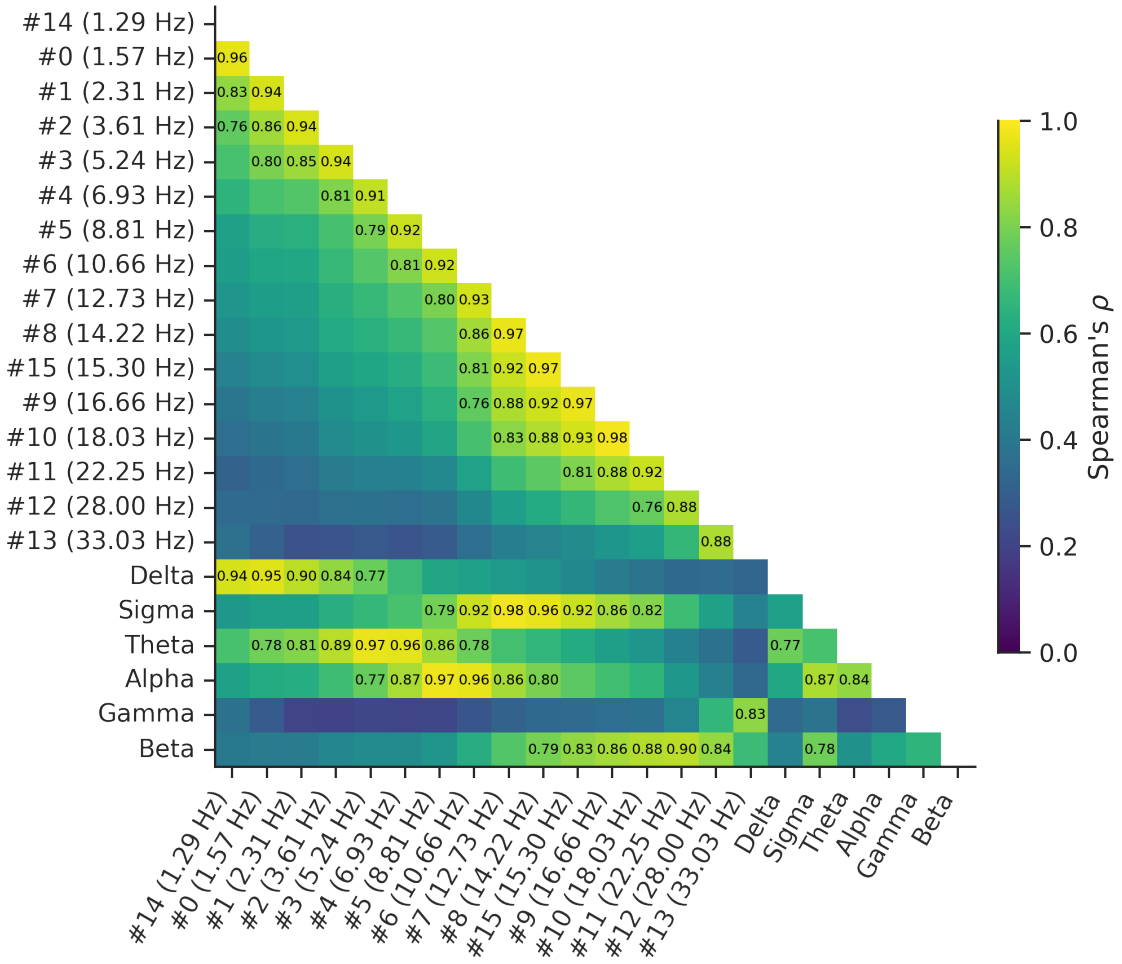


Figure S2: Heatmap showing Spearman correlation coefficients (ρ) between whole-night average power computed using conventional canonical bands and power computed using the parameters from the 16 trained Gabor filters ("AI-derived filters"). The block-diagonal pattern indicates that distinct AI-derived filters correlate with particular physiological bands (e.g., Gabor filters #0–#2 with the delta band and #7–#8 with the sigma/alpha bands). This pattern is consistent with functional alignment between the learned filters and classical neurophysiological bands.

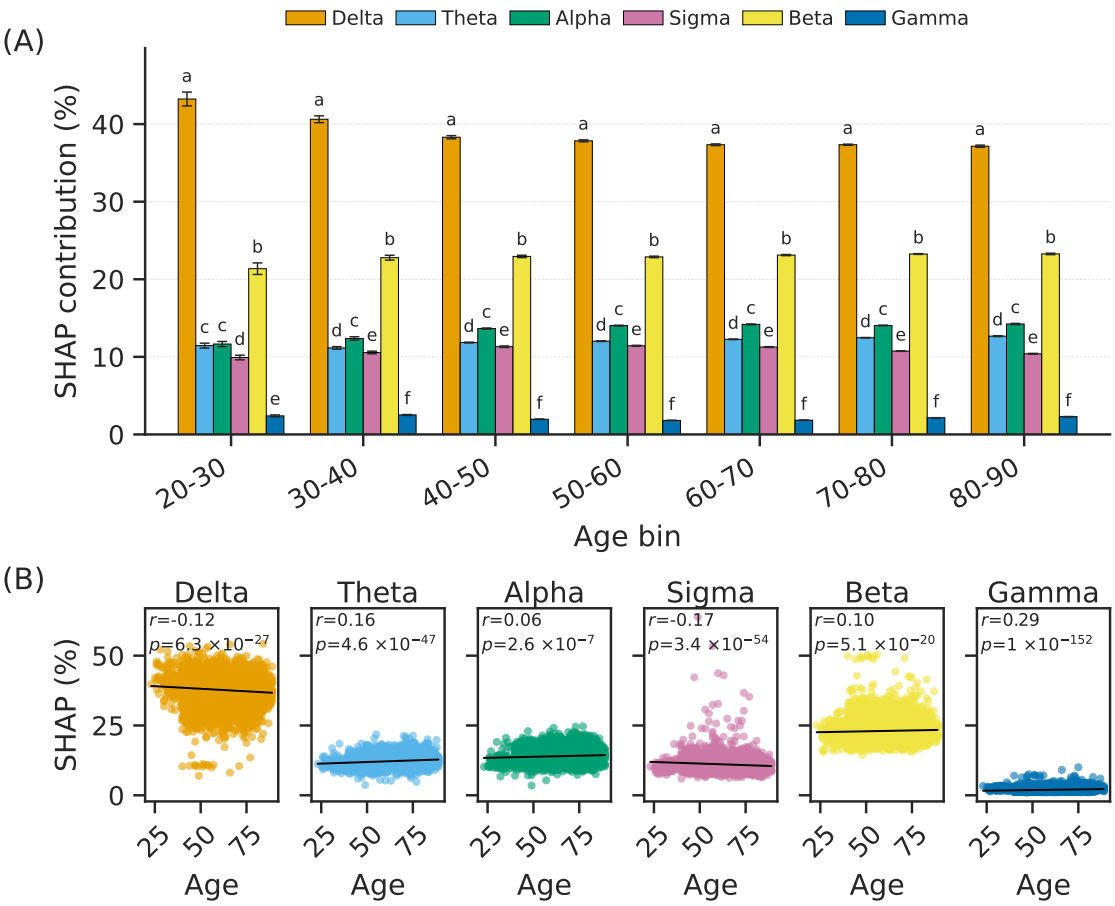


Figure S3: The model's reliance on different EEG frequency bands changes with age during **NREM** sleep. (A) The mean absolute SHAP contribution (%) for each AI-derived EEG band, grouped by 10-year age bins. Error bars represent the 95% confidence interval of the mean. A generalized linear model confirmed a significant interaction between brain bands and age group ($\chi^2(30) = 1866.5$, $p < 0.001$). Letters above the bars denote statistical significance based on post-hoc pairwise comparisons within each age group, adjusted with a Bonferroni-Holm correction. Bars that do not share a common letter are significantly different ($p < 0.05$). (B) Scatter plots showing the relationship between age and SHAP contribution for each band across all individuals. For robust statistical inference, monotonic trends were assessed using Spearman's rank correlation (r), with the corresponding p-value displayed. For intuitive visualization of the trend, an ordinary least squares (OLS) regression line is overlaid.

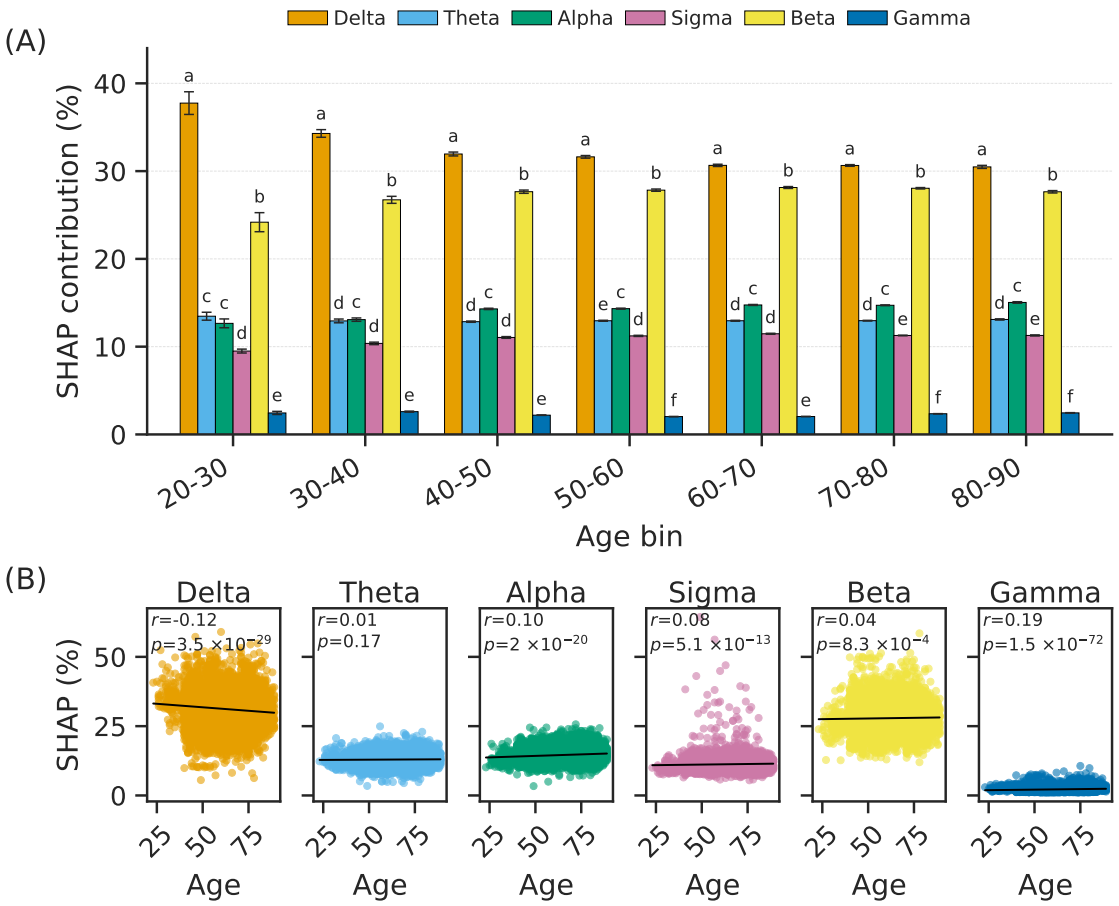


Figure S4: The model's reliance on different EEG frequency bands changes with age during **REM** sleep. (A) The mean absolute SHAP contribution (%) for each AI-derived EEG band, grouped by 10-year age bins. Error bars represent the 95% confidence interval of the mean. A generalized linear model confirmed a significant interaction between brain bands and age group ($\chi^2(30) = 929.7$, $p < 0.001$). Letters above the bars denote statistical significance based on post-hoc pairwise comparisons within each age group, adjusted with a Bonferroni-Holm correction. Bars that do not share a common letter are significantly different ($p < 0.05$). (B) Scatter plots showing the relationship between age and SHAP contribution for each band across all individuals. For robust statistical inference, monotonic trends were assessed using Spearman's rank correlation (r), with the corresponding p-value displayed. For intuitive visualization of the trend, an ordinary least squares (OLS) regression line is overlaid.

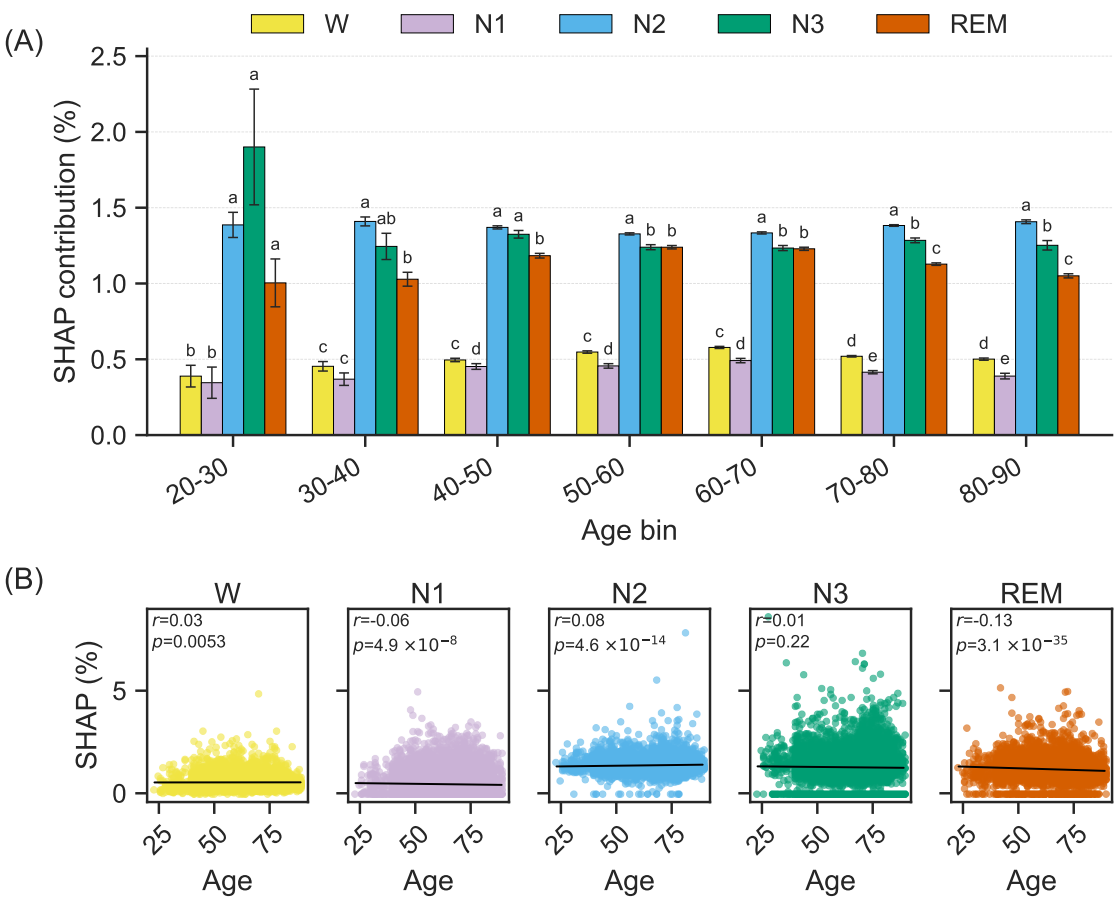


Figure S5: The model's reliance on different sleep stages, analyzed using SHAP contributions normalized by stage duration. Each value represents the relative importance of a given sleep stage per 5-minute epoch. (A) Mean duration-normalized SHAP contribution (%) for each sleep stage, grouped by 10-year age bins. Error bars represent the 95% confidence interval of the mean. A generalized linear model confirmed a significant interaction between sleep stage and age group ($\chi^2(24) = 335.3$, $p < 0.001$). Letters above the bars denote statistical significance from post-hoc pairwise comparisons within each age group (adjusted with a Bonferroni-Holm correction); bars that do not share a common letter are significantly different ($p < 0.05$). (B) Scatter plots showing the relationship between chronological age and the duration-normalized SHAP contribution for each sleep stage. Monotonic trends were assessed using Spearman's rank correlation (r_s), with corresponding p-values displayed. An ordinary least squares (OLS) regression line is overlaid on each plot for visualization.

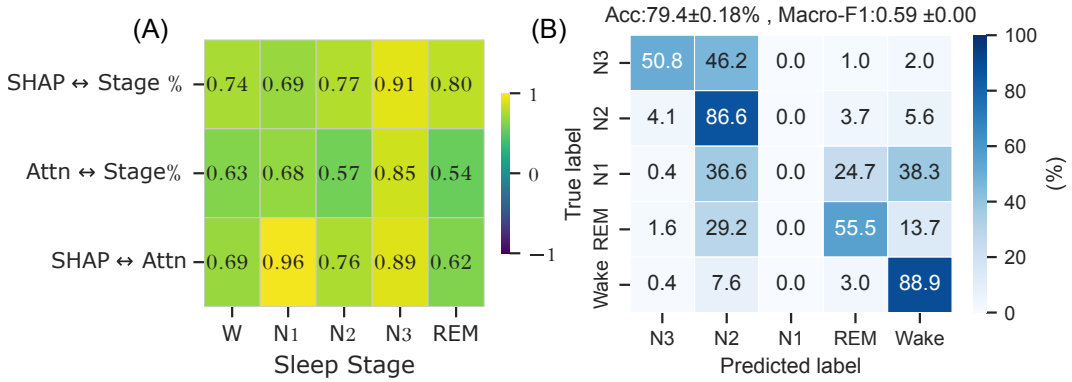


Figure S6: Analysis of sleep-stage information captured by the model. (A) Correlation between model attribution scores and ground-truth sleep-stage prevalence. This panel displays Spearman correlation coefficients between three sets of values: model-derived SHAP values, internal attention weights (Attn), and the ground-truth percentage of time spent in each sleep stage (Stage %). The top row shows the correlation between SHAP values and Stage %, while the bottom row shows the correlation between SHAP values and attention weights. (B) Performance of a linear probe for sleep staging using the model's learned embeddings. A logistic regression classifier was trained to predict sleep stages from the embeddings of test set participants (n=8,629). Performance was evaluated using a five-fold cross-validation across subjects. The confusion matrix shows results pooled from all test folds to provide an overall evaluation of the classifier. The N1 sleep stage is underrepresented in this analysis due to our epoch-labeling methodology. We assigned a single sleep stage label to a 5-minute epoch only if that stage constituted more than 60% of the epoch. Because N1 sleep is typically fragmented, it seldom met this criterion to become the dominant stage.

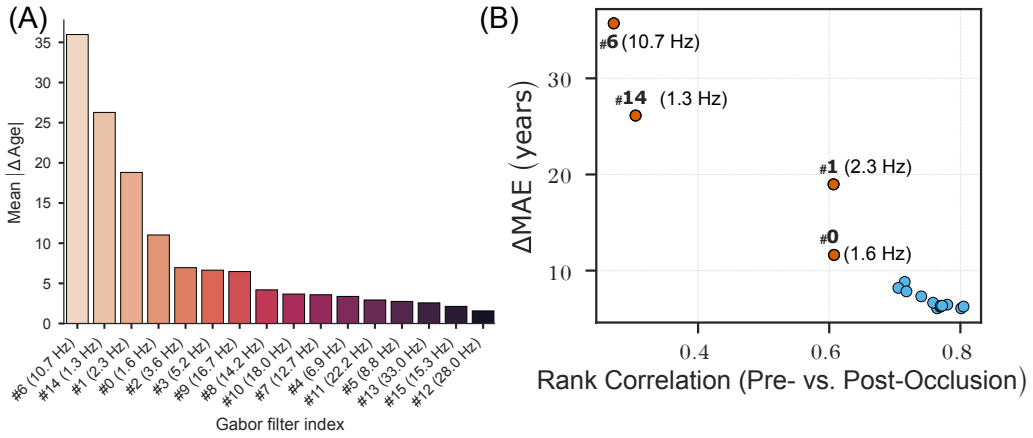


Figure S7: Per-filter occlusion analysis of the learned Gabor filters. (A) The change in model performance when each Gabor filter is individually occluded. Filters are ranked along the x-axis according to their impact on the model's mean absolute error (△MAE), which is shown on the y-axis. This analysis was conducted on the internal test set ($n = 8,629$). The ranking highlights the influence of specific filters discussed in the text, particularly those in the delta range (e.g., #0, #1, and #14) and near the alpha/sigma boundary (e.g., #6 at ∼10.7 Hz). (B) Relationship between the change in prediction error and the rank-order stability of predictions after occluding each filter. The y-axis represents △MAE, while the x-axis represents the Spearman rank correlation between the model's age predictions before and after occlusion. This analysis serves as a sanity check; however, it does not account for interactions between filters.

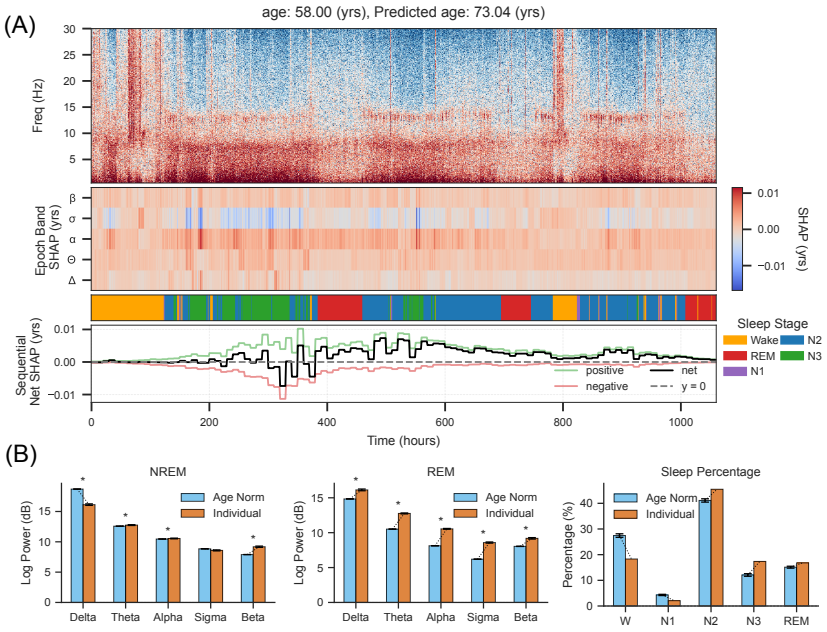


Figure S8: (A) Full-night analysis of a 58-year-old subject predicted to be 73.0 years old, including (from top to bottom) the spectrogram, epoch-level band SHAP contributions, expert-scored hypnogram, and the net sequential SHAP contribution. The sequential SHAP analysis (bottom track) identifies the initial period of consolidated N3 sleep (e.g., hours 1.5-3.5) as contributing to a 'younger-age' prediction (negative values), while the later fragmented sleep shows sustained positive values. Concurrently, the spectral-band-level SHAP analysis (middle track) shows that sigma power provides a consistent negative ('younger-age') contribution throughout the night. (B) Quantitative comparison of the individual's sleep metrics (orange bars) against an age-matched normative mean (blue bars; * denotes $p < 0.05$, Mann-Whitney U test). The subject shows a greater proportion of N2 and N3 sleep and less wakefulness than the age-matched norm. However, delta power during NREM sleep is lower than the norm, which is consistent with the model's overall older-age estimate. In contrast, the subject's sigma power is comparable to the norm, consistent with its negative SHAP contribution in Panel A. This example illustrates spectral-band-level and sleep-stage-level readouts and is not intended as an individualized clinical explanation.

4 Supplementary Tables

Table S1: Mean Absolute Error (MAE) by Age Group for Test and External Validation Sets.

Test Set (n=8629)			**External Validation (n=190)**		
Age	n	MAE	Age	n	MAE
20-25	1	2.48	20-25	6	3.35 ± 1.40
25-30	19	7.00 ± 5.78	25-30	8	6.48 ± 5.85
30-35	69	5.73 ± 5.20	30-35	17	6.62 ± 4.53
35-40	52	5.06 ± 5.47	35-40	31	5.52 ± 4.22
40-45	421	6.41 ± 5.24	40-45	24	3.64 ± 3.92
45-50	509	5.92 ± 5.46	45-50	29	5.85 ± 5.23
50-55	690	5.56 ± 4.75	50-55	29	4.15 ± 4.22
55-60	986	5.93 ± 4.66	55-60	25	6.27 ± 4.61
60-65	814	6.78 ± 4.83	60-65	9	5.44 ± 4.20
65-70	1070	6.61 ± 4.78	65-70	3	5.37 ± 5.33
70-75	1719	6.29 ± 4.89	70-75	8	7.40 ± 4.91
75-80	1399	5.08 ± 5.04	75-80	0	–
80-85	723	4.46 ± 5.48	80-85	1	15.49
85-90	157	5.95 ± 4.71	85-90	0	–
Average		5.66 ± 1.16			6.30 ± 3.14

Table S2: **Cohort-stratified age-prediction performance.**

Cohort	n	Age (years)	MAE (years)	Pearson r
SHHS	5,183	62.86 ± 10.65	6.23 ± 5.19	0.725
MrOS	2,588	75.86 ± 5.01	5.07 ± 4.43	0.393
CFS	131	50.20 ± 10.28	6.11 ± 5.28	0.790
STAGES	727	49.78 ± 11.43	6.38 ± 5.28	0.744
Internal test set overall	8,629	65.47 ± 12.20	5.89 ± 5.01	0.811
HomePAP external validation	190	46.11 ± 12.04	5.41 ± 4.57	0.840

Values are reported as mean ± SD unless otherwise specified. MAE, mean absolute error. Pearson r denotes the correlation between predicted and chronological age. The internal test set combines SHHS, MrOS, CFS, and STAGES; HomePAP was evaluated as an external validation cohort.

Table S3: Summary of Frequency Analysis with EEG Subbands

#	f (Hz)	Init. f (Hz)	σ	$\triangle f$ (Hz)	Corresponding EEG subband
#0	1.57	1.0	1.18	0.32	Delta
#1	2.31	2.0	1.76	0.21	Delta
#2	3.61	3.0	3.06	0.12	Delta
#3	5.24	5.0	3.09	0.12	Theta
#4	6.93	7.0	3.34	0.11	Theta
#5	8.81	9.0	4.07	0.09	Alpha
#6	10.66	11.0	3.94	0.09	Alpha
#7	12.73	13.0	2.58	0.14	Alpha
#8	14.22	14.5	1.97	0.19	Beta
#9	16.66	16.0	1.86	0.20	Beta
#10	18.03	18.0	1.51	0.25	Beta
#11	22.25	22.0	0.03	10.75	Beta
#12	28.00	28.0	0.06	6.30	Beta
#13	33.03	32.0	0.07	5.57	Gamma
#14	1.29	0.5	2.39	0.16	Delta
#15	15.30	15.0	1.29	0.29	Beta

Each learnable Gabor filter is defined by two parameters: its center frequency f and bandwidth parameter σ. $\triangle f$ is the -3 dB bandwidth. The filters were initialized with different starting center frequencies (Init. f, Hz) and a fixed bandwidth parameter of $\sigma = 0.25$. This initial σ value corresponds to a $\triangle f$ of approximately 1.5 Hz.

Table S4: **Cohort-stratified all-background spectral attribution trends with age.**

Cohort	Band	n	Spearman ρ	p value	Mean contribution (%)
SHHS	Delta	5182	-0.158	< 0.001	33.41
	Theta	5182	0.111	< 0.001	11.83
	Alpha	5182	0.142	< 0.001	14.35
	Sigma	5182	-0.120	< 0.001	11.92
	Beta	5182	0.116	< 0.001	26.37
	Gamma	5182	0.084	< 0.001	2.12
MrOS	Delta	2588	-0.151	< 0.001	31.51
	Theta	2588	0.059	0.00276	10.92
	Alpha	2588	0.070	< 0.001	13.69
	Sigma	2588	-0.067	< 0.001	10.86
	Beta	2588	0.084	< 0.001	29.41
	Gamma	2588	0.146	< 0.001	3.62
CFS	Delta	131	-0.194	0.026	35.39
	Theta	131	0.149	0.089	11.02
	Alpha	131	0.089	0.311	13.37
	Sigma	131	-0.006	0.950	10.52
	Beta	131	0.132	0.132	26.64
	Gamma	131	0.090	0.307	3.06
STAGES	Delta	727	-0.239	< 0.001	34.89
	Theta	727	-0.033	0.369	11.12
	Alpha	727	0.190	< 0.001	13.39
	Sigma	727	0.065	0.078	11.00
	Beta	727	0.181	< 0.001	26.60
	Gamma	727	0.106	0.00437	2.99
Overall	Delta	8628	-0.277	< 0.001	32.99
	Theta	8628	-0.039	< 0.001	11.48
	Alpha	8628	0.057	< 0.001	14.05
	Sigma	8628	-0.170	< 0.001	11.50
	Beta	8628	0.298	< 0.001	27.31
	Gamma	8628	0.373	< 0.001	2.66

Mean contribution denotes the percentage of normalized absolute all-background spectral SHAP contribution assigned to each frequency band. Spearman ρ quantifies the association between chronological age and band-level contribution within each cohort. Cohort membership was identified from the recording filename. The overall analysis includes internal test subjects with available SHAP files. SHAP, Shapley additive explanations.

Table S5: **Cohort-stratified NREM spectral attribution trends with age.**

Cohort	Band	n	Spearman ρ	p value	Mean contribution (%)
SHHS	Delta	5182	-0.121	< 0.001	37.45
	Theta	5182	0.186	< 0.001	12.44
	Alpha	5182	0.076	< 0.001	14.31
	Sigma	5182	-0.153	< 0.001	11.60
	Beta	5182	0.121	< 0.001	22.54
	Gamma	5182	0.124	< 0.001	1.66
MrOS	Delta	2588	-0.076	< 0.001	37.96
	Theta	2588	0.121	< 0.001	12.26
	Alpha	2588	0.042	0.033	13.78
	Sigma	2588	-0.139	< 0.001	10.81
	Beta	2588	0.074	< 0.001	22.87
	Gamma	2588	0.142	< 0.001	2.32
CFS	Delta	131	-0.114	0.196	40.69
	Theta	131	0.262	0.00251	11.66
	Alpha	131	-0.006	0.942	13.24
	Sigma	131	-0.137	0.118	10.92
	Beta	131	0.109	0.217	21.39
	Gamma	131	0.105	0.234	2.10
STAGES	Delta	719	-0.221	< 0.001	38.16
	Theta	719	0.053	0.156	11.36
	Alpha	719	0.196	< 0.001	13.18
	Sigma	719	-0.007	0.859	11.13
	Beta	719	0.172	< 0.001	23.62
	Gamma	719	0.124	< 0.001	2.55
Overall	Delta	8620	-0.105	< 0.001	37.71
	Theta	8620	0.164	< 0.001	12.28
	Alpha	8620	0.043	< 0.001	14.04
	Sigma	8620	-0.186	< 0.001	11.32
	Beta	8620	0.124	< 0.001	22.71
	Gamma	8620	0.284	< 0.001	1.94

Mean contribution denotes the percentage of normalized absolute NREM spectral SHAP contribution assigned to each frequency band. Spearman ρ quantifies the association between chronological age and band-level NREM contribution within each cohort. NREM includes N1, N2, and N3 epochs. Cohort membership was identified from the recording filename. The overall analysis includes internal test subjects with available SHAP files and usable NREM epochs. SHAP, Shapley additive explanations.

Table S6: Multivariable associations between clinical covariates and brain age gap

Covariate	β (95% CI)	p value
Chronological age, per year	-0.231 (-0.248, -0.215)	< 0.001
Male vs female	2.183 (1.783, 2.584)	< 0.001
BMI, per 5 kg/m^2	-0.086 (-0.297, 0.125)	0.424
AHI, per 10 events/h	0.001 (-0.141, 0.143)	0.987
Arousal index, per 10 events/h	0.686 (0.487, 0.885)	< 0.001
Stroke history	0.580 (-0.360, 1.520)	0.227
Depression history	0.737 (0.014, 1.460)	0.046
CHD history	0.484 (-0.011, 0.978)	0.055
Diabetes history	0.752 (0.148, 1.355)	0.015
Hypertension history	0.603 (0.244, 0.961)	< 0.001
Smoking history	-1.058 (-1.842, -0.275)	0.008

Note. Coefficients represent associations with brain age gap in years. Confidence intervals and p values are based on HC3 robust standard errors. The model included sex, BMI, AHI, arousal index, and the listed medical-history variables.

Table S7: Cox model results for all-cause mortality and CVD mortality on MrOS and SHHS dataset

Covariate	CVD Mortality		All-cause Mortality	
	HR (95% CI)	p	HR (95% CI)	p
Age (years)	1.17 (1.15–1.19)	< 0.001	1.13 (1.12–1.14)	< 0.001
Gender (Female)	0.70 (0.54–0.90)	0.01	0.71 (0.63–0.81)	< 0.001
Race (Non-White)	1.20 (0.88–1.64)	0.25	0.99 (0.84–1.16)	0.86
Body mass index (BMI, kg$\cdot$ m^{-2})	1.00 (0.98–1.03)	0.78	1.00 (0.99–1.02)	0.72
Smoking status (current or past)	1.54 (0.97–2.43)	0.07	1.91 (1.57–2.33)	< 0.001
Hypertension	1.33 (1.07–1.65)	0.01	1.25 (1.13–1.37)	< 0.001
Diabetes	2.05 (1.60–2.64)	< 0.001	1.49 (1.31–1.71)	< 0.001
CHD	2.22 (1.76–2.81)	< 0.001	1.49 (1.32–1.67)	< 0.001
Stroke	1.26 (0.87–1.83)	0.22	1.25 (1.03–1.53)	0.02
Depression	1.25 (0.85–1.84)	0.26	1.47 (1.24–1.73)	< 0.001
AHI (events/hour)	1.00 (1.00–1.01)	0.40	1.00 (1.00–1.01)	0.06
ArI (events/hour)	0.99 (0.98–1.00)	0.09	0.99 (0.99–1.00)	0.03
BAG (per 10 years)	1.25 (1.07–1.46)	0.01	1.19 (1.11–1.28)	< 0.001

The mortality analysis was performed with (n = 6,891, deaths = 1,820) for all-cause mortality and (n = 6,891, deaths = 415) for cardiovascular mortality. HR, hazard ratio; CI, confidence interval; BAG, brain age gap; CHD, coronary heart disease; AHI, apnea-hypopnea index; ArI, arousal index.

Table S8: Cox model results for all-cause mortality and CVD mortality on SHHS dataset

Covariate	CVD Mortality		All-cause Mortality	
	HR (95% CI)	p	HR (95% CI)	p
Age (years)	1.17 (1.14–1.19)	< 0.001	1.12 (1.11–1.13)	< 0.001
Gender (Female)	0.70 (0.53–0.91)	0.0071	0.73 (0.64–0.83)	< 0.001
Race (Non-White)	1.02 (0.70–1.49)	0.92	0.93 (0.75–1.14)	0.49
Body mass index (BMI, kg$\cdot$ m^{-2})	1.00 (0.97–1.03)	0.90	1.00 (0.98–1.01)	0.58
Smoking status (current or past)	1.90 (1.18–3.07)	0.0084	1.99 (1.59–2.48)	< 0.001
Hypertension	1.63 (1.22–2.17)	< 0.001	1.37 (1.20–1.57)	< 0.001
Diabetes	2.58 (1.87–3.56)	< 0.001	1.77 (1.47–2.13)	< 0.001
CHD	2.18 (1.57–3.03)	< 0.001	1.46 (1.20–1.78)	< 0.001
Stroke	1.42 (0.92–2.20)	0.12	1.31 (1.03–1.68)	0.03
Depression	1.18 (0.69–2.00)	0.55	1.40 (1.11–1.77)	0.0044
AHI (events/hour)	1.00 (0.99–1.01)	0.62	1.00 (1.00–1.01)	0.11
ArI (events/hour)	0.99 (0.97–1.00)	0.14	0.99 (0.99–1.00)	0.17
BAG (per 10 years)	1.15 (0.96–1.37)	0.13	1.17 (1.07–1.28)	< 0.001

The mortality analysis was performed with cohort data. HR, hazard ratio; CI, confidence interval; BAG, brain age gap; CHD, coronary heart disease; AHI, apnea-hypopnea index; ArI, arousal index.

Table S9: Cox model results for all-cause mortality and CVD mortality on MrOS dataset

Covariate	CVD Mortality		All-cause Mortality	
	HR (95% CI)	p	HR (95% CI)	p
Age (years)	1.17 (1.13–1.22)	< 0.001	1.15 (1.13–1.17)	< 0.001
Race (Non-White)	1.58 (0.91–2.77)	0.11	1.10 (0.84–1.45)	0.47
Body mass index (BMI, kg$\cdot$m^{-2})	1.03 (0.98–1.08)	0.27	1.02 (1.00–1.04)	0.12
Smoking status (current or past)	0.00 (0.00–0.00)	< 0.001	1.56 (0.99–2.47)	0.06
Hypertension	0.97 (0.69–1.35)	0.84	1.12 (0.97–1.29)	0.13
Diabetes	1.27 (0.84–1.94)	0.26	1.25 (1.03–1.51)	0.02
CHD	2.42 (1.74–3.37)	< 0.001	1.51 (1.31–1.74)	< 0.001
Stroke	0.85 (0.37–1.95)	0.70	1.13 (0.81–1.57)	0.48
Depression	1.33 (0.75–2.37)	0.33	1.51 (1.18–1.93)	0.0011
AHI (events/hour)	1.00 (0.99–1.02)	0.65	1.00 (1.00–1.01)	0.41
ArI (events/hour)	0.99 (0.97–1.01)	0.46	0.99 (0.99–1.00)	0.087
BAG (per 10 years)	1.60 (1.12–2.28)	0.0092	1.27 (1.11–1.46)	< 0.001

The mortality analysis was performed on the MrOS cohort with data. HR, hazard ratio; CI, confidence interval; BAG, brain age gap; CHD, coronary heart disease; AHI, apnea-hypopnea index; ArI, arousal index.

Table S10: Stage-stratified spectral regression analysis comparing canonical, even-cutpoint, and learned Gabor band-power features during NREM and REM sleep.

Model Variant	Cross-validated R^2 (95% CI)	Adjusted R^2	ΔR^2	p-value
	NREM			
Model 1 (Canonical band powers)	0.447 (0.433–0.461)	0.449	–	–
Model 2 (Even-cutpoint band powers, n=16)	0.450 (0.437–0.464)	0.454	+0.003	0.080
Model 3 (Aggregated learned Gabor band powers)	0.467 (0.453–0.482)	0.468	+0.020	5.66×10^{-4}
Model 4 (Learned Gabor band powers, n=16)	0.505 (0.489–0.520)	0.507	+0.058	5.44×10^{-7}
Model 5 (Canonical + even-cutpoint bands, n=16)	0.461 (0.442–0.480)	0.467	+0.014	0.025
Model 6 (Canonical + aggregated learned Gabor bands)	0.493 (0.471–0.514)	0.501	+0.045	6.30×10^{-5}
Model 7 (Canonical + learned Gabor bands, n=16)	0.519 (0.487–0.550)	0.534	+0.072	1.23×10^{-4}
	REM			
Model 1 (Canonical band powers)	0.424 (0.409–0.439)	0.425	–	–
Model 2 (Even-cutpoint band powers, n=16)	0.427 (0.413–0.442)	0.430	+0.003	0.069
Model 3 (Aggregated learned Gabor band powers)	0.424 (0.409–0.439)	0.425	+0.000	0.947
Model 4 (Learned Gabor band powers, n=16)	0.436 (0.422–0.450)	0.438	+0.012	1.01×10^{-4}
Model 5 (Canonical + even-cutpoint bands, n=16)	0.431 (0.417–0.446)	0.434	+0.007	1.89×10^{-3}
Model 6 (Canonical + aggregated learned Gabor bands)	0.437 (0.421–0.452)	0.439	+0.013	1.29×10^{-5}
Model 7 (Canonical + learned Gabor bands, n=16)	0.451 (0.437–0.466)	0.455	+0.027	1.48×10^{-6}

Prediction performance was evaluated with 10-fold stratified cross-validation within each sleep-stage stratum. Folds were stratified by sex, body-mass index, 5-year age bins, and cohort identifier. Cross-validated R^2 was computed on the held-out 10% subset within each fold, and adjusted R^2 was computed on the full sample as a complementary predictor-count-penalized metric. ΔR^2 and p-values compare each model with the stage-specific Model 1 using paired t-tests on fold-wise R^2 values. The even-cutpoint bands used 16 contiguous frequency bands over 0.5–40 Hz and therefore matched the number of learned Gabor band-power features. In direct matched comparisons, learned Gabor band powers (n=16) outperformed even-cutpoint band powers (n=16) during NREM ($\Delta R^2 = 0.0544$, paired fold-wise $p = 5.66 \times 10^{-7}$) and REM ($\Delta R^2 = 0.0086$, paired fold-wise $p = 1.66 \times 10^{-3}$). Because several predictors were collinear (variance inflation factor, VIF > 10), individual regression coefficients were not interpreted.

Table S11: Bandwise agreement between age correlations obtained with classical and AI-derived spectral features

Band	ρ_{CLS} (95% CI)	ρ_{AI} (95% CI)	p
	Full night		
Delta	−0.312 (−0.330, −0.293)	−0.279 (−0.299, −0.258)	0.060
Theta	−0.204 (−0.223, −0.184)	−0.145 (−0.165, −0.125)	0.001
Alpha	−0.100 (−0.121, −0.080)	−0.067 (−0.086, −0.045)	0.075
Sigma	−0.201 (−0.220, −0.181)	−0.206 (−0.225, −0.186)	1.000
Beta	0.082 (0.061, 0.102)	0.079 (0.058, 0.100)	1.000
Gamma	0.252 (0.233, 0.273)	0.294 (0.274, 0.315)	0.015
	NREM		
Delta	−0.287 (−0.306, −0.268)	−0.245 (−0.265, −0.224)	0.012
Theta	−0.170 (−0.189, −0.150)	−0.138 (−0.159, −0.117)	0.140
Alpha	−0.181 (−0.202, −0.162)	−0.163 (−0.183, −0.142)	0.450
Sigma	−0.331 (−0.348, −0.311)	−0.353 (−0.371, −0.334)	0.276
Beta	−0.176 (−0.198, −0.155)	−0.190 (−0.211, −0.171)	0.450
Gamma	0.072 (0.052, 0.092)	0.109 (0.088, 0.130)	0.070
	REM		
Delta	−0.144 (−0.164, −0.123)	−0.147 (−0.167, −0.127)	1.000
Theta	−0.118 (−0.139, −0.098)	−0.115 (−0.136, −0.094)	1.000
Alpha	−0.048 (−0.068, −0.028)	−0.038 (−0.057, −0.016)	1.000
Sigma	−0.077 (−0.099, −0.056)	−0.082 (−0.103, −0.060)	1.000
Beta	−0.127 (−0.148, −0.107)	−0.127 (−0.146, −0.106)	1.000
Gamma	0.057 (0.037, 0.077)	0.054 (0.033, 0.074)	1.000
	Wake		
Delta	−0.240 (−0.258, −0.221)	−0.216 (−0.236, −0.197)	0.300
Theta	−0.138 (−0.158, −0.118)	−0.103 (−0.124, −0.083)	0.100
Alpha	−0.101 (−0.122, −0.079)	−0.058 (−0.080, −0.035)	0.024
Sigma	−0.148 (−0.168, −0.128)	−0.124 (−0.146, −0.104)	0.300
Beta	−0.038 (−0.059, −0.019)	−0.004 (−0.027, 0.017)	0.100
Gamma	0.157 (0.138, 0.178)	0.177 (0.158, 0.198)	0.300

Spearman correlations between chronological age and band power were computed for canonical bands (ρ_{CLS}) and AI-derived bands (ρ_{AI}). 95% confidence intervals were estimated from 2000 bootstrap resamples. Reported p-values are Holm-adjusted values for comparisons between the classical and AI-derived correlations.

Table S12: Summary statistics across datasets

Variable	SHHS	MrOS	CFS	STAGES	HomePAP
N	5698	2855	612	1464	190
Train (n, %)	461 (8.09%)	240 (8.41%)	434 (70.92%)	662 (45.2%)	0 (0%)
Val (n, %)	54 (0.95%)	27 (0.95%)	47 (7.68%)	75 (5.12%)	0 (0%)
Test (n, %)	5183 (91.0%)	2588 (90.65%)	131 (21.41%)	727 (49.66%)	190 (100%)
Age (years)	63.13 ±11.12	76.14 ±5.23	46.77 ±16.35	46.04 ±14.30	46.11 ±12.0
BMI (kg/m²)	28.18 ±5.09	27.21 ±3.80	34.03 ±9.14	31.18 ±8.79	36.93 ±9.04
Sex (female)	2981 (52.3%)	0 (0%)	337 (55.1%)	767 (52.4%)	99 (52.11%)
Systolic BP (mmHg)	127.40 ±19.33	126.77 ±16.29	125.40 ±15.99	–	124.84 ±14.12
Diastolic BP (mmHg)	73.73 ±11.60	67.73 ±9.43	74.95 ±9.17	–	76.13 ±10.72
AHI (events/hr)	17.99 ±16.15	21.29 ±16.25	14.59 ±17.83	–	14.0 ±18.5
Arousal Index (/hr)	19.17 ±10.65	23.59 ±11.67	16.82 ±10.33	–	23.8 ±18.1
Sleep Efficiency (%)	82.78 ±10.54	76.16 ±11.94	77.13 ±12.87	80.9 ±11.6	78.2 ±14.9
Total Sleep Time (min)	360.09 ±64.36	355.87 ±69.07	366.66 ±76.00	352.3 ±80.1	335.5 ±75.3
REM Sleep (%)	19.84 ±6.25	19.28 ±6.65	18.77 ±7.45	19.1 ±13.2	18.05 ±8.21
N1 Sleep (%)	5.43 ±3.95	6.86 ±4.36	5.31 ±4.08	11.9 ±10.6	15.72 ±12.5
N2 Sleep (%)	56.51 ±11.71	62.75 ±9.65	58.46 ±11.57	57.9 ±26.9	52.7 ±13.3
N3 Sleep (%)	18.22 ±11.88	11.24 ±9.00	17.29 ±10.55	11.06 ±10.6	13.6 ±9.68

AHI, apnea–hypopnea index; BP, blood pressure (mmHg); BMI, body mass index.

Table S13: Phase 1: Epoch-level Feature Encoder and Short-Sequence Aggregator.

Layer #	Layer Type	Parameters / Properties
Input: 10 consecutive 30-second EEG epochs (C3, C4), shape: $(10, 2, 30 \times F_s)$		
1–9	**Per-Epoch CNN Encoder**	Processes each of the 10 epochs independently:
1	Gabor Layer	$N_{\text{kernels}} = 16$, Stride = 1, Length = $2 \times F_s$
2	1D Conv Block I	$N_{\text{filters}} = 32$, Kernel Size = 3, Stride = 2
3	1D Conv Block II	$N_{\text{filters}} = 64$, Kernel Size = 3, Stride = 2
4	1D Conv Block III	$N_{\text{filters}} = 128$, Kernel Size = 3, Stride = 2
5–7	1D Conv Block IV (×3)	$N_{\text{filters}} = 256$, Kernel Size = 3, Stride = 2
8	1D Adaptive Avg. Pooling	Reduces temporal dimension to 1
9	Fully Connected	$N_{\text{neurons}} = 128$. Output for 10 epochs: (10, 128)
10	**Additive Attention**	Aggregates the 10 epoch embeddings into a single vector.
11	*Fully Connected*	*Regression head for age estimation*
Output: Single 128-dimensional embedding for the 5-minute window.		

This model processes 10 consecutive 30-second EEG epochs (a 5-minute window). The encoder contains six convolutional layers. Each convolutional layer is followed by batch normalization → ELU activation → layer normalization. $F_s = 128$ Hz.

Table S14: Phase 2: Full-Night Sequence Model.

Layer #	Layer Type	Parameters / Properties
Input: Sequence of L segment embeddings, shape: (L, 128)		
1	Bidirectional LSTM	Layers = 1, Hidden Dimension = 128
2	Additive Attention	Aggregates LSTM outputs into a single 256-dim context vector
3	Fully Connected	$N_{\text{neurons}} = 128$
4	Dropout	$p = 0.5$
5	Fully Connected	$N_{\text{neurons}} = 1$
Output: Age estimate		

This model processes the sequence of 5-minute window embeddings generated by the Phase 1 model to produce a final, night-level prediction. LSTM: long short-term memory.

Table S15: Age-stratified spectral SHAP contributions across pooled cohorts.

	Mean absolute SHAP contribution (%)					
Age (years)	Delta	Theta	Alpha	Sigma	Beta	Gamma
20–30	38.49 ± 5.40	11.53 ± 1.55	12.23 ± 1.76	10.27 ± 1.28	24.60 ± 4.79	2.88 ± 0.94
30–40	36.59 ± 4.67	11.08 ± 1.67	12.69 ± 1.94	10.95 ± 1.59	25.80 ± 3.86	2.89 ± 0.72
40–50	34.47 ± 5.81	11.44 ± 1.55	13.75 ± 2.02	11.78 ± 2.48	26.17 ± 4.53	2.39 ± 0.76
50–60	34.27 ± 5.00	11.54 ± 1.60	13.97 ± 2.07	11.96 ± 2.16	26.06 ± 4.08	2.21 ± 0.70
60–70	33.36 ± 4.57	11.62 ± 1.62	14.26 ± 2.15	11.78 ± 2.07	26.67 ± 3.68	2.31 ± 0.72
70–80	32.01 ± 4.34	11.41 ± 1.82	14.08 ± 2.30	11.20 ± 1.68	28.31 ± 3.80	2.99 ± 0.97
80–90	31.08 ± 4.29	11.42 ± 1.89	14.26 ± 2.30	10.94 ± 1.45	28.98 ± 3.83	3.32 ± 0.98

Values are mean ± SD percentages of normalized absolute spectral SHAP contribution assigned to each frequency band.